Chats With Legal Application

Course Name: GenAI

Medicaps University – Datagami Skill Based Course

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Academic Year: 2025-26

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

Results & Output

Add the below details here:

1. Screenshots / outputs
2. Reports / dashboards / models
3. Key outcomes

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

1. Problem Statement & Objectives

1.1 Problem Statement

In today's digital environment, accessing and understanding legal information remains a significant challenge for common users. Legal documents are typically written in highly technical language, which makes them difficult to interpret without proper legal knowledge. As a result, individuals often struggle to understand their rights, obligations, and legal procedures.

This issue becomes more critical when users need quick answers but are unable to consult legal professionals due to time and cost constraints. Therefore, there is a strong need for a system that can bridge this gap by simplifying legal knowledge and making it accessible in an easy-to-understand format.

Key Problems Identified:

- Complex legal terminology difficult for non-experts
- Limited access to affordable legal guidance
- Time-consuming consultation process
- Lack of awareness about legal rights

1.2 Project Objectives

The objective of this project is to design and develop an AI-powered legal chatbot that can assist users in understanding legal concepts in a

simplified manner. The system leverages Generative AI to provide human-like responses and enhance user interaction.

Main Objectives:

- Develop an intelligent chatbot using Generative AI
- Simplify complex legal language into understandable responses
- Provide instant answers to user queries
- Improve accessibility of legal information
- Demonstrate real-world application of AI technologies

1.3 Scope of the Project

The scope of the project defines the boundaries and capabilities of the system. This application focuses on providing general legal assistance through conversational AI.

Scope Includes:

- Answering general legal queries
- Providing simplified explanations
- Real-time conversational interaction
- Supporting students and general users

Limitations:

- Not a replacement for professional legal advice
- Limited to available data and model capabilities
- May not handle complex legal cases accurately

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2. Proposed Solution

2.1 Key Features

The system is designed with multiple features to enhance usability and performance.

Key Features:

- AI-powered legal chatbot
- Natural Language Processing (NLP)
- Context-aware responses
- Real-time query handling
- Simplified legal explanations
- User-friendly interface
- Scalable backend system
- Optional RAG integration for accuracy

2.2 Overall Architecture / Workflow

The system follows a structured architecture where each component plays a vital role in processing user queries and generating responses.

Workflow Explanation:

The process begins when a user enters a legal query into the chatbot interface. The frontend captures this input and sends it to the backend server. The backend processes the request, formats it into a suitable prompt, and forwards it to the LLM API.

The LLM analyzes the query and generates a response based on its training data and context. If RAG is implemented, relevant documents

are retrieved from the vector database to enhance accuracy. Finally, the generated response is sent back to the frontend and displayed to the user.

Steps Involved:

- User enters query
- Frontend sends request
- Backend processes input
- LLM generates response
- Response returned to user

2.3 Tools & Technologies Used

Various tools and technologies are used to build and deploy the system efficiently.

Technologies Used:

- **Programming Language:** Python
- **Backend Framework:** LangChain
- **Frontend:** Streamlit
- **LLM API:** Mistral
- **Vector Database:** FAISS
- **Embedding :** Hugging Face
- **Version Control:** Git & GitHub
- **API Testing:** Postman

Each of these technologies plays an important role in ensuring smooth system performance and scalability.

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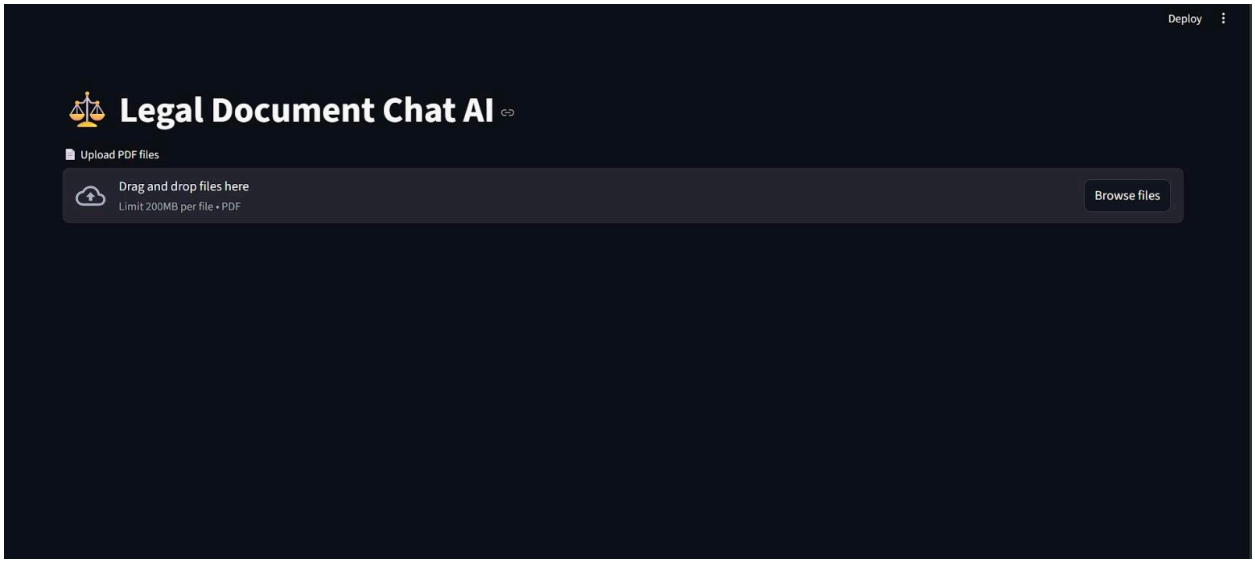
3. Results & Output

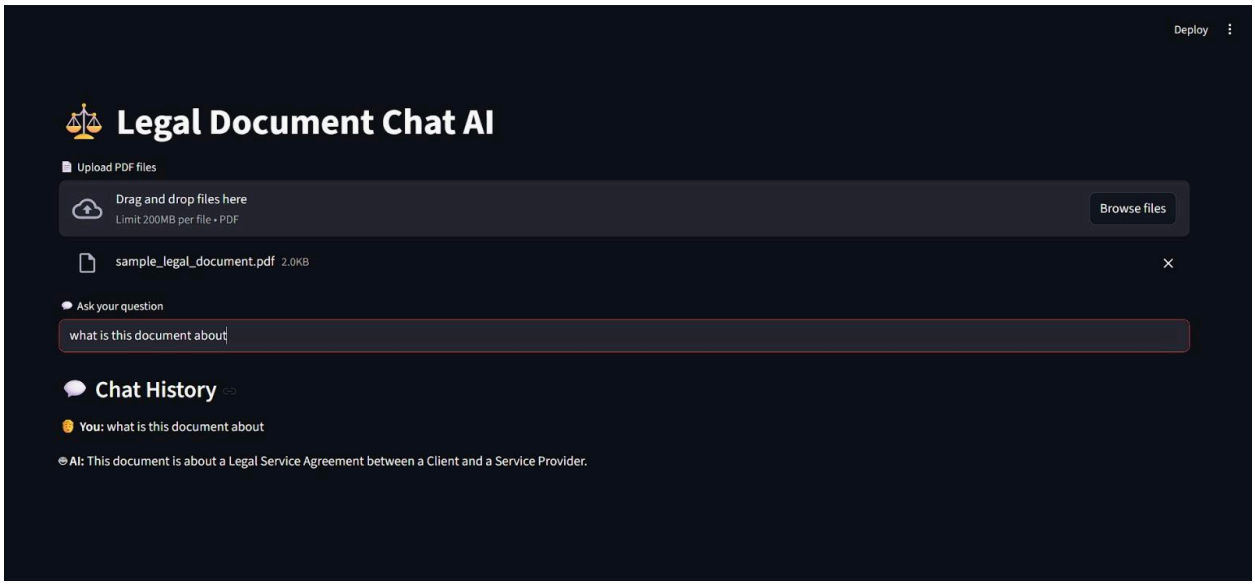
3.1 Screenshots / Outputs

The results of the project can be demonstrated through screenshots and outputs. These provide visual proof of system functionality.

Include the Following:

- Chat interface
- User query example
- AI-generated response
- Backend API response





3.2 Reports / Dashboards / Models

The system generates various outputs that can be analyzed to evaluate performance.

Reports Include:

- Chat logs of user interactions
- Prompt-response mapping
- Model output samples
- Performance observations

3.3 Key Outcomes

The project successfully achieves its objectives by delivering a functional AI-based chatbot system.

Key Outcomes:

- Developed a working legal chatbot
- Enabled real-time AI responses
- Simplified legal knowledge for users
- Improved accessibility and usability
- Demonstrated AI integration in legal domain

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4. Conclusion

The Gen AI Based Legal Chat Application demonstrates how advanced AI technologies can be used to simplify complex domains like law. The chatbot provides quick, clear, and user-friendly responses, making legal knowledge more accessible.

Throughout this project, significant learning was achieved in multiple areas.

Key Learnings:

- Understanding Generative AI concepts
- Working with LLM APIs
- Backend development using FastAPI
- Prompt engineering techniques
- System design and architecture

This project highlights the potential of AI in transforming traditional systems into intelligent solutions.

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5. Future Scope & Enhancements

The system can be further improved by adding more advanced features and capabilities.

Future Enhancements:

- Integration with real legal databases
- Multi-language support
- Voice-based chatbot
- Document analysis (contracts, agreements)
- Mobile application development
- Fine-tuned legal AI models
- Integration with government systems

Additional Content for Expansion

Use Case Scenarios

The system can be applied in various real-world scenarios where users need quick legal assistance.

Examples:

- Understanding tenant rights
- Learning legal concepts
- Checking contract terms

Sample Queries

Examples:

- What are my legal rights?
- Explain breach of contract
- What is FIR?

Learning Outcomes

This project helped in gaining both technical and conceptual knowledge.

Skills Developed:

- AI application development
- API integration
- Problem-solving skills
- System design understanding